

Lab 38: End-to-End Mini ML Pipeline

Aim

To build a complete mini machine-learning pipeline that loads a dataset, cleans and preprocesses it using Pandas, trains a KNN classifier, and reports predictions.

Algorithm

1. Read the dataset into a Pandas DataFrame.
2. Handle missing values.
3. Encode categorical labels into numerical values.
4. Split the dataset into training and testing sets.
5. Train a KNN classifier.
6. Predict the classes for the test data.
7. Display the predictions and accuracy.

Program

```
# Lab 38: End-to-End Mini ML Pipeline
```

```
import pandas as pd

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Load Iris dataset into a Pandas DataFrame
iris = load_iris()

df = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
```

```
)
```

```
df["target"] = iris.target
```

```
# Check for missing values
```

```
print("Missing Values:\n", df.isnull().sum())
```

```
# Features and target
```

```
X = df.iloc[:, :-1]
```

```
y = df["target"]
```

```
# Split the dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
```

```
    test_size=0.2,
```

```
    random_state=42
```

```
)
```

```
# Train KNN classifier
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X_train, y_train)
```

```
# Make predictions
```

```
y_pred = model.predict(X_test)
```

```
print("\nPredictions:", y_pred)
```

```
print("Actual Values:", y_test.values)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Sample Output

Missing Values:

sepal length (cm) 0

sepal width (cm) 0
petal length (cm) 0
petal width (cm) 0
target 0
dtype: int64

Predictions: [1 0 2 1 1 ...]

Actual Values: [1 0 2 1 1 ...]

Accuracy: 1.0

Result

Thus, an **end-to-end mini machine learning pipeline was successfully implemented**, including loading the dataset, checking/cleaning the data, training a KNN classifier, and predicting the output values.